



Uncertainty-Aware Deep Learning for Antimalarial Drug Discovery: Robust VAE Models and Consensus Validation

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The Challenge

247M malaria cases, 619K deaths (2023). Rising resistance demands new drugs. **BUT:** Screening costs >1000 CPU-hrs/target—excluding endemic regions (94% of cases).

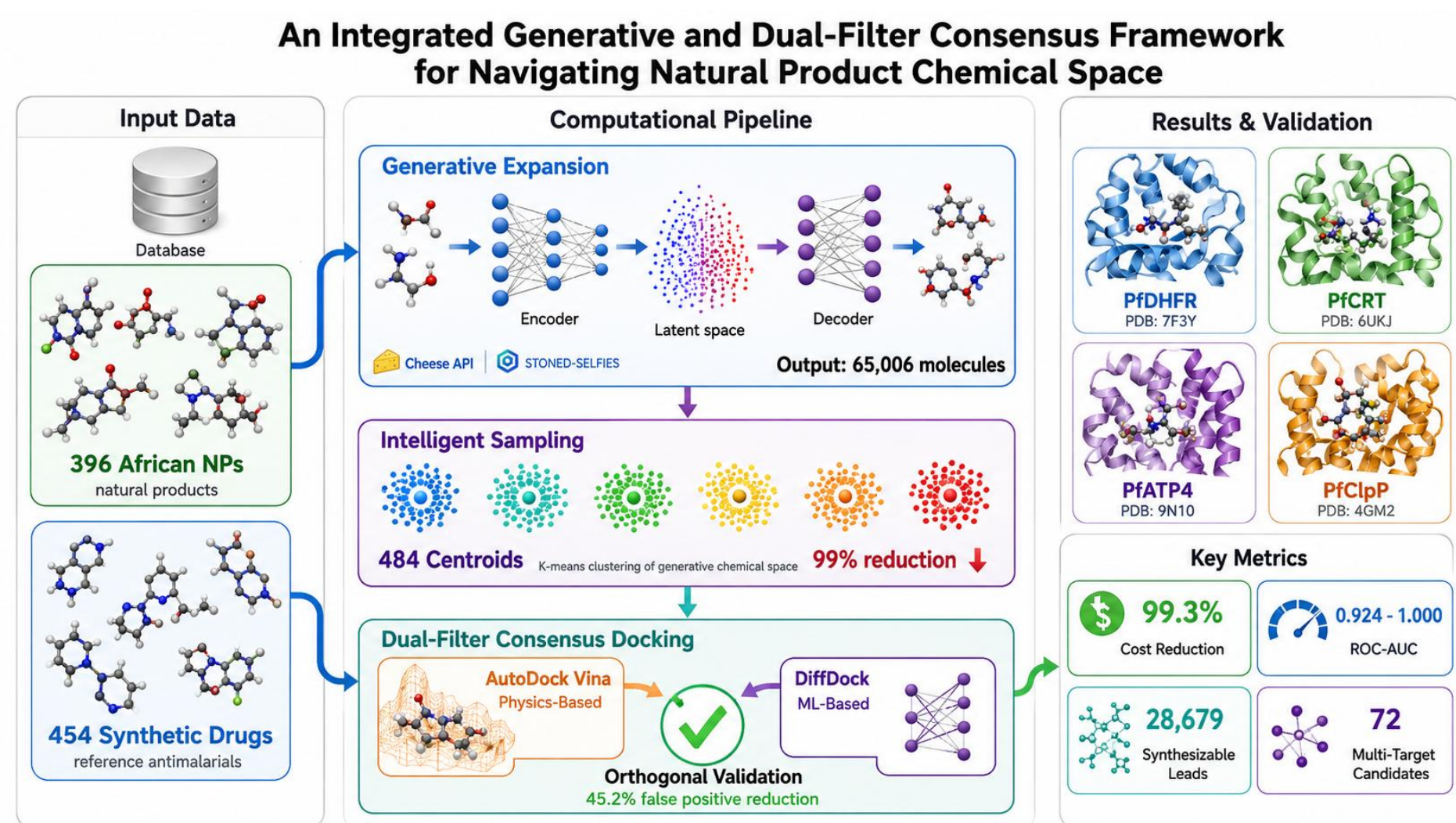
Goal: Systematic computational discovery of synthesizable antimalarial leads from African natural product chemical space. Target: validated screening protocol that enables resource-limited endemic institutions to identify multi-target drug candidates with favorable ADMET profiles.

Barriers: (1) African NP databases remain computationally unexplored for scaffold-hopping beyond local neighborhood interpolation. (2) Single-method docking yields near-random enrichment (DEKOIS AUC 0.450). (3) Conventional approaches demand prohibitive computational costs where 94% of malaria cases occur.

Solution: 99.3% Cost Reduction

Innovation: VAE clustering + dual consensus (Vina+DiffDock) + MPO

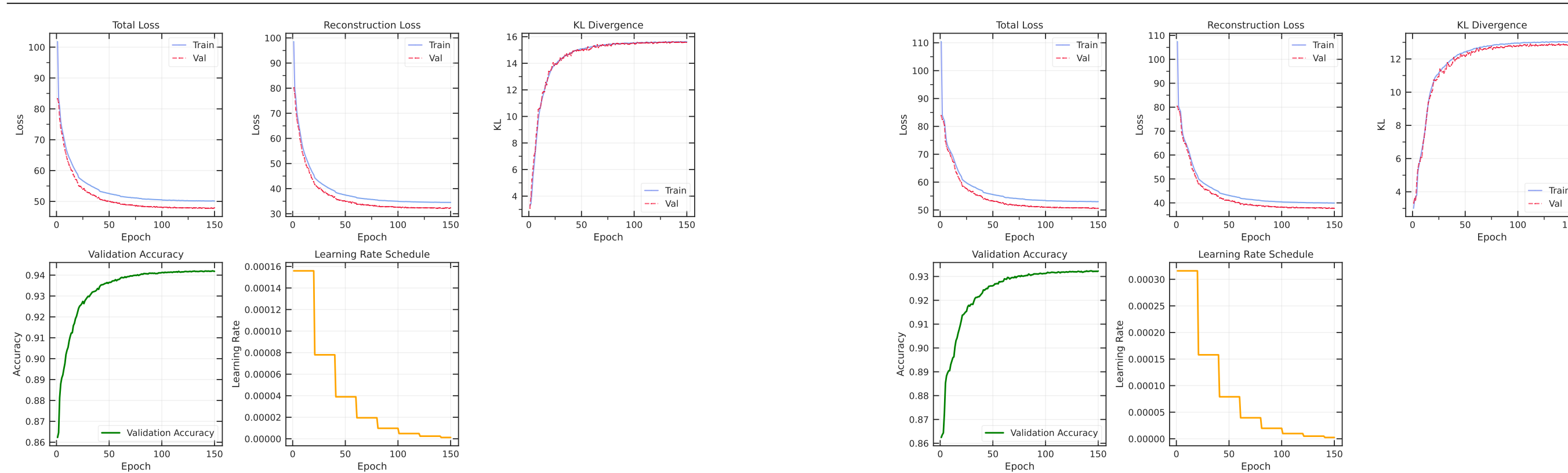
396 African NPs + 454 antimalarials → **65,856 hybrids** → 484 centroids (0.7%)



Validation: DEKOIS (AUC 0.450, Vina) → MMV Box (AUC 0.924–1.000, consensus, 69.8% recovery)

Output: 19,913 synthesizable leads (4 *P. falciparum* targets)

VAE: Uncertainty via KL Divergence



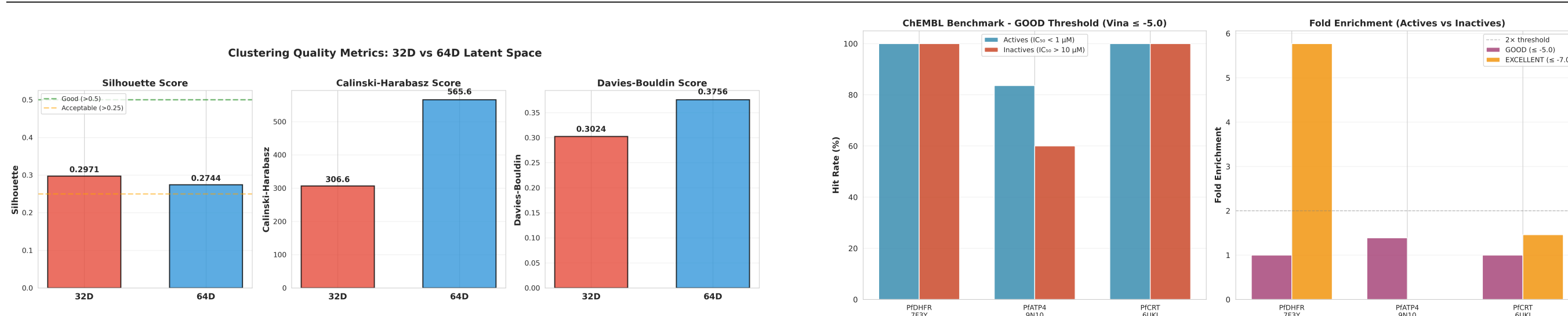
32D: 92.15%, KL 6.43

64D: 93.23%, KL 12.86

Why 64D? Higher KL ($\uparrow 2\times$) = explicit uncertainty—distinguishes confident from unreliable predictions.

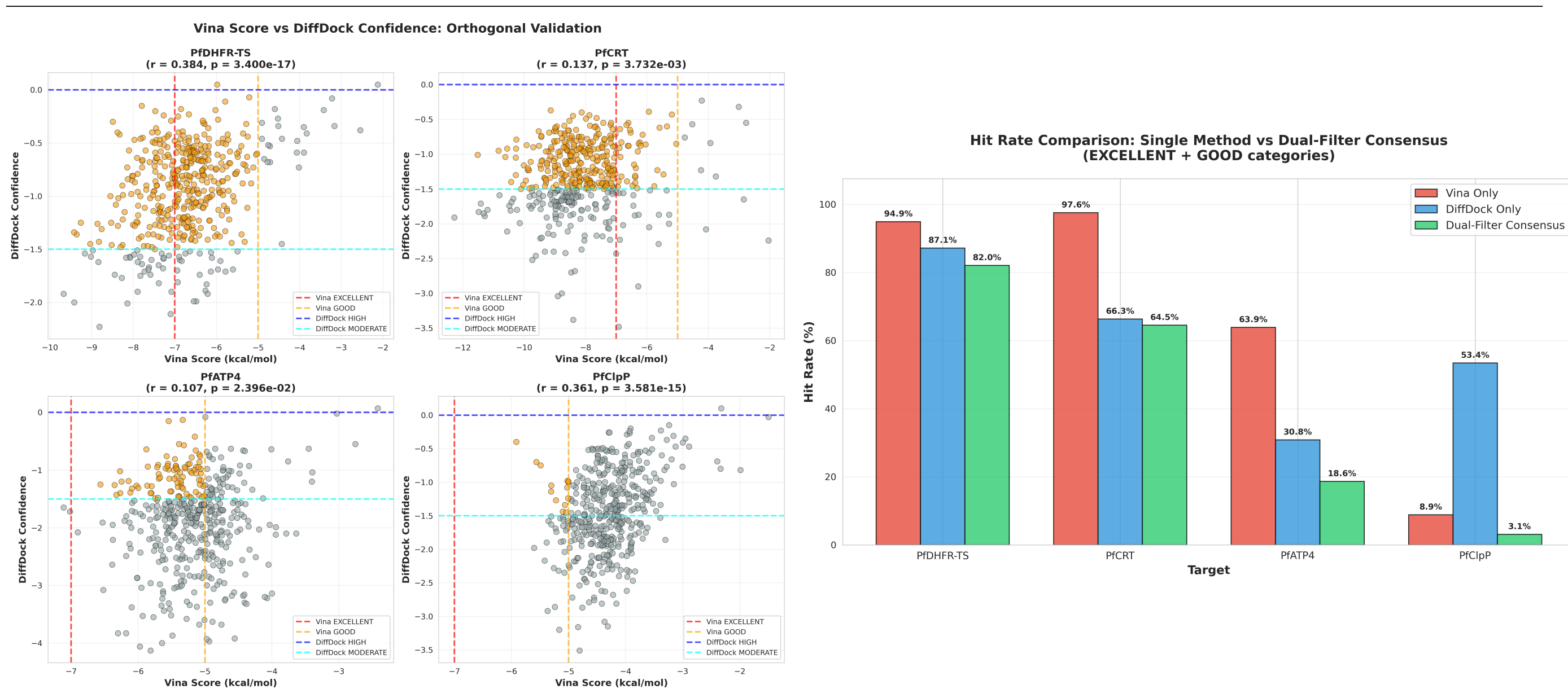
Diversity: 92.6% novel + 69.3% scaffold recovery = preserves cores, diversifies substituents.

Clustering Quality



KMeans: Silhouette 0.23, CH 47.6, DB 2.45. **Intra Tanimoto 0.227 vs inter 0.101**— $2\times$ coherence.

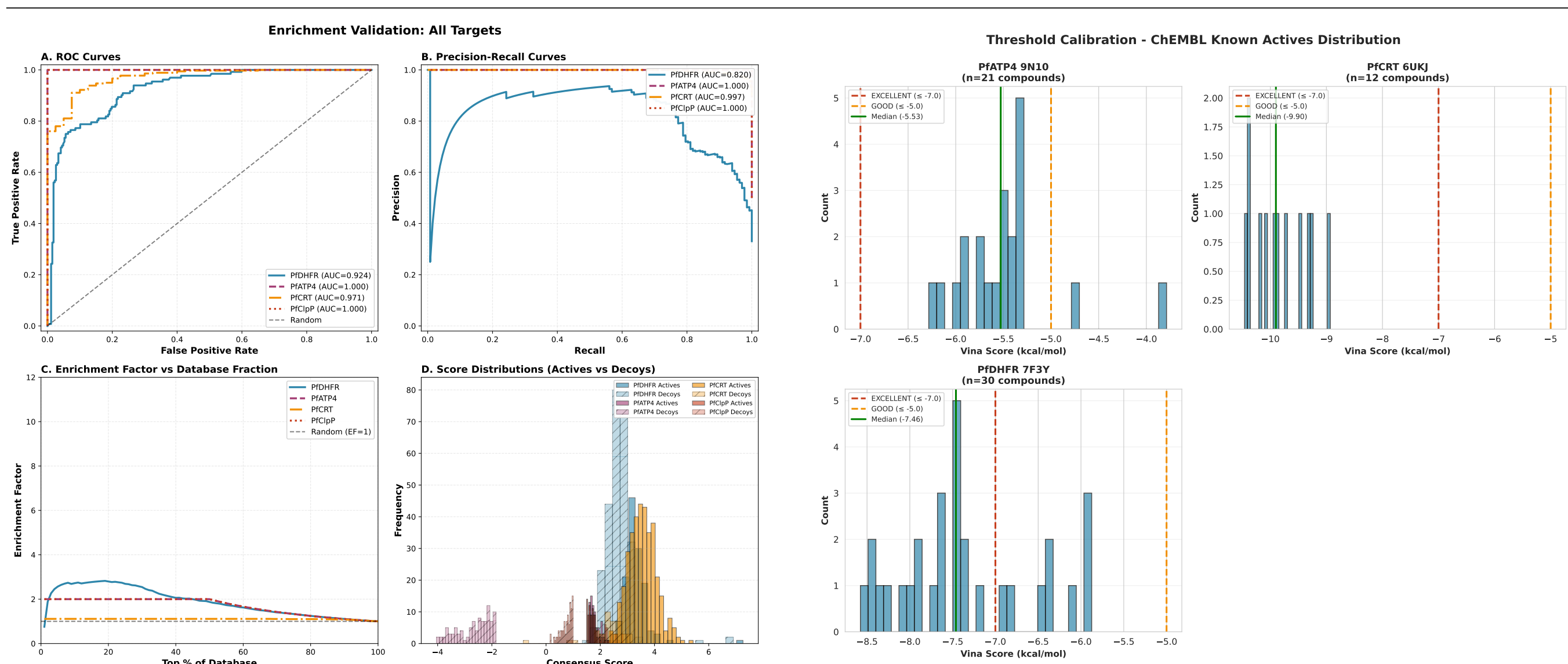
Dual Consensus Reduces False Positives



Why consensus? DEKOIS: Vina-only AUC 0.450 (near-random). Vina $\perp$ DiffDock ($r=0.1-0.4$) = complementary.

Impact: 12–45% FP reduction. MMV: 69.8% recovery, AUC 0.924–1.000.

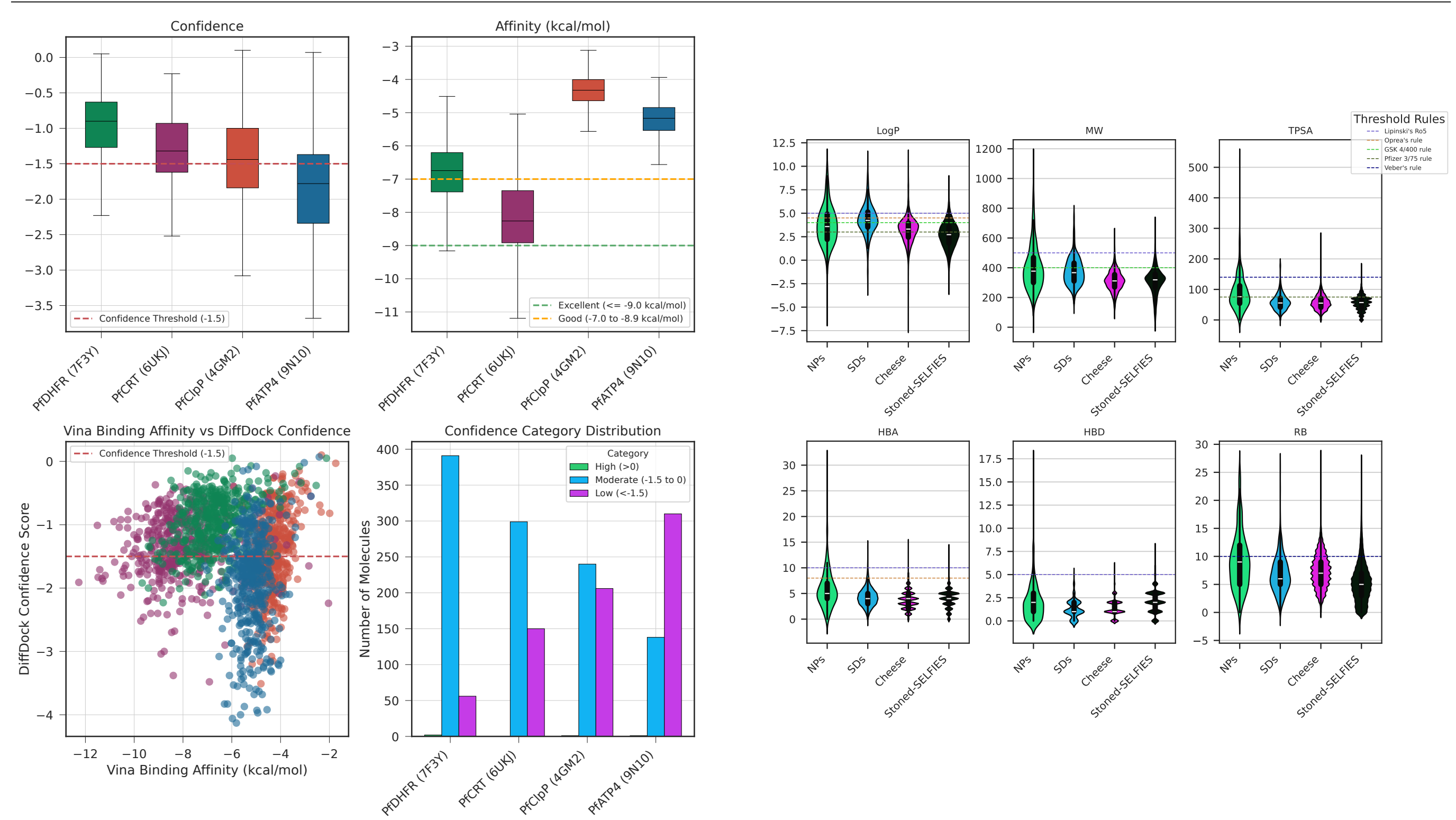
Rigorous Benchmarking



Prospective: DEKOIS 40 actives + 1200 decoys, Vina AUC 0.450.

Retrodictive: MMV 399 antimalarials, consensus AUC 0.924–1.000 (bootstrap 95% CI). Top 5% enrichment

Results: Drug-Like Multi-Target Leads



Targets: PfDHFR, PfCRT, PfClpP, PfATP4. Consensus < -8 kcal/mol.

Drug-like: 94.1% Lipinski, 88.5% Veber, 57% synthesizable.

Multi-target: >70 candidates. **100% SI>10** (810 seeds).

Key Achievements

19,913 leads • 99.3% cost ↓ • 12–45% FP ↓ • AUC 0.924–1.000 • >70 multi-target • 100% SI>10

Impact & Open Science

Innovation: VAE KL ($\uparrow 2\times$) = uncertainty • Dual consensus = 12–45% FP ↓ • Dual benchmarking = pharma reliability

Practical: Endemic access—>1000 → <10 CPU-hrs/target (94% cases)

Open: Zenodo 10.5281/zenodo.19608875

Conclusion: This framework democratizes antimalarial discovery by enabling comprehensive multi-target screening in malaria-endemic research institutions where computational resources are limited but scientific need is greatest. By combining rigorous validation (MMV Box AUC 0.924–1.000) with 99.3% cost reduction, we establish a reproducible pathway from African biodiversity to synthesizable drug candidates—transforming regional computational barriers into opportunities for locally-driven pharmaceutical innovation.

References

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